

Answers

1. 3	2. 3	3. 1	4. D	5. B	6. C	7. B	8. C
9. A	10. B	11. B	12. B	13. D	14. B	15. B	16. C
17. D	18. C	19. C	20. B	21. E	22. A	23. C	24. E
25. D	26. C	27. B	28. D	29. B	30. D	31. C	32. E
33. A	34. C	35. D	36. C	37. D	38. C	39. C	40. C
41. A							

Explanations

1.3

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

For a question created by anyone among Amal, Bimal, Komal a question can be reviewed after one review or two reviews.

The information provided states :

Q02, Q06, Q09, Q11, and Q12 were rejected. Q01, Q03, Q04, Q05, Q07, Q08, Q10, Q13 were accepted.

For the questions :

Amal - Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.

Bimal - Q02, Q04, Q06, Q07, Q08, Q09, Q12, and Q13.

Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

Q02-is reviewed by Amal, Bimal, and Komal and was rejected this has a possibility of being prepared by an external person and faced rejection by one among Amal and Bimal and Komal. (SME).

Q03-Was accepted and reviewed by Amal, Komal and hence must have been prepared by Bimal(Bimal).

Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).

Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)

Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).

Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)

Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)

Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Amal definitely prepared questions Q07, Q09, Q12.

2.3

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

For a question created by anyone among Amal, Bimal, Komal a question can be reviewed after one review or two reviews.

The information provided states :

Q02, Q06, Q09, Q11, and Q12 were rejected. Q01, Q03, Q04, Q05, Q07, Q08, Q10, Q13 were accepted.

For the questions :

Amal - Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.

Bimal - Q02, Q04, Q06, Q07, Q08, Q09, Q12, and Q13.

Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

Q02-is reviewed by Amal, Bimal, and Komal and was rejected this has a possibility of being prepared by an external person and faced rejection by one among Amal and Bimal and Komal. (SME).

Q03-Was accepted and reviewed by Amal, Komal and hence must have been prepared by Bimal(Bimal).

Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).

Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)

Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).

Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)

Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)

Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Q02, Q04, Q08 were prepared SME

3.1

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

For a question created by anyone among Amal, Bimal, Komal a question can be reviewed after one review or two reviews.

The information provided states :

Q02, Q06, Q09, Q11, and Q12 were rejected. Q01, Q03, Q04, Q05, Q07, Q08, Q10, Q13 were accepted.

For the questions :

Amal - Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.

Bimal - Q02, Q04, Q06, Q07, Q08, Q09, Q12, and Q13.

Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

Q02-is reviewed by Amal, Bimal, and Komal and was rejected this has a possibility of being prepared by an external person and faced rejection by one among Amal and Bimal and Komal. (SME).

Q03-Was accepted and reviewed by Amal, Komal and hence must have been prepared by Bimal(Bimal).

Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).

Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)

Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).

Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)

Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)

Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Q10 was definitely prepared by Komal.

 VIDEO SOLUTION

4.D

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

For a question created by anyone among Amal, Bimal, Komal a question can be reviewed after one review or two reviews.

The information provided states :

Q02, Q06, Q09, Q11, and Q12 were rejected. Q01, Q03, Q04, Q05, Q07, Q08, Q10, Q13 were accepted.

For the questions :

Amal - Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.

Bimal - Q02, Q04, Q06, Q07, Q08, Q09, Q12, and Q13.

Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

Q02-is reviewed by Amal, Bimal, and Komal and was rejected this has a possibility of being prepared by an external person and faced rejection by one among Amal and Bimal and Komal. (SME).

Q03-Was accepted and reviewed by Amal, Komal and hence must have been prepared by Bimal(Bimal).

Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).

Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)

Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).

Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)

Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)

Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Amal reviewed the questions : Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.

For Q2 - Amal can either approve or reject the question.

For Q3- Amal must approve the question.

For Q4 -Amal can either approve or reject the question.

For Q06 - Amal must reject the question.

For Q08 - Amal can either approve or reject the question.

For Q10 - Amal must approve the question.

For Q11 - Amal must reject the question.

For Q13- Amal can either approve or reject the question.

The approval ratio has a minimum of 2 questions only Q3, Q10, and a maximum of 6 questions Q02, Q3, Q04, Q08, Q10, Q13.

Hence $2/8 = 0.25$ to $6/8 = 0.75$

 VIDEO SOLUTION

5. B

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

For a question created by anyone among Amal, Bimal, Komal a question can be reviewed after one review or two reviews.

The information provided states :

Q02, Q06, Q09, Q11, and Q12 were rejected. Q01, Q03, Q04, Q05, Q07, Q08, Q10, Q13 were accepted.

For the questions :

Amal - Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.

Bimal - Q02, Q04, Q06, Q07, Q08, Q09, Q12, and Q13.

Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

Q02-is reviewed by Amal, Bimal, and Komal and was rejected this has a possibility of being prepared by an external person and faced rejection by one among Amal and Bimal and Komal. (SME).

Q03-Was accepted and reviewed by Amal, Komal and hence must have been prepared by Bimal(Bimal).

Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).

Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)

Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).

Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)

Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)

Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Q06 irrespective of the question prepared by Komal/SME Bimal must reject the question.

Q07, Q09, Q12 are prepared by Amal and were reviewed by Bimal first and Komal next for this to happen Bimal must reject these questions.

Hence a total of four questions were rejected by Bimal.

 VIDEO SOLUTION



6. C

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

For a question created by anyone among Amal, Bimal, Komal a question can be reviewed after one review or two reviews.

The information provided states :

Q02, Q06, Q09, Q11, and Q12 were rejected. Q01, Q03, Q04, Q05, Q07, Q08, Q10, Q13 were accepted.

For the questions :

Amal - Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.

Bimal - Q02, Q04, Q06, Q07, Q08, Q09, Q12, and Q13.

Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

Q02-is reviewed by Amal, Bimal, and Komal and was rejected this has a possibility of being prepared by an external person and faced rejection by one among Amal and Bimal and Komal. (SME).

Q03-Was accepted and reviewed by Amal, Komal and hence must have been prepared by Bimal(Bimal).

Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).

Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)

Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).

Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)

Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)

Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Amal could have possibly created the questions Q07, Q09, Q12 of which all three of them were reviewed by two people each, and hence they must have been rejected by the first reviewer.

Bimal could have created the questions :

Q01, Q03, Q05, Q 11.

Q01, Q05 are reviewed only by Komal and hence did not have any rejection.

Q03, Q11 faced at least one rejection.

Hence a total of five questions.

 VIDEO SOLUTION

7. B

Let us calculate the minimum average that Venkat can obtain in this case, and for this to happen, one and a half times the expected results must be obtained for company B. The value is calculated as follows,

	A	B	C	D	AVERAGE
Before results	20	10	30	40	25
After results	20	15	60	40	33.75

We can see that the minimum average obtained in the case is 33.75%, and Venkat cannot earn less than 33.75% in this case, making statement 2 true.

As we can see above, in the case of a 33.75% average return, Company A did not report extraordinary results, and Company C must belong to an auto or steel company, as it received one and a half times the expected results. So, statement 3 is false and 4 is true.

For the average to be maximum, one and a half must be obtained from company D.

$$\text{Average maximum return} = \frac{20 + 10 + 60 + 60}{4} = 37.5\%$$

We can see that statement 1 is also false.

So, the correct statements are 2 and 4 only.

Hence, the correct answer is option B.

[VIEW SOLUTION](#)

8. C

For the average overall to be 38.75%, the sum of all the results has to be $38.75 \times 4 = 155\%$. The sum is 155 in only a single case, when the results of D are doubled and the results of C become one and a half times the expected results. The calculations are as shown in the table below,

	A	B	C	D	Average
Before results	20	10	30	40	25
After results	20	10	45	80	38.75

To give a 38.75 % average return, we can see that B didn't give extraordinary returns, and we know that one of the auto or steel industries had one and a half times the expected results. As Company C got one and a half times the expected results, Company C has to be either the auto or steel industry.

So, the correct statements are 1 and 4 only.

Hence, the correct answer is option C.

[VIEW SOLUTION](#)

9. A

For the average profit to be minimum, the companies with returns higher than the initially expected returns have to be the companies with the least expected returns.

So, the company with twice the initially expected returns has to be the company that was expected to return 10%. So, the returns now become 20%.

The company with one and a half times the initially expected returns has to be the company with the next least expected results, which is 20%. So, the returns now become 30%.

The companies with 30% and 40% expected results got the same as the expected results.

The average overall profit can be calculated as,

$$\frac{20 + 30 + 30 + 40}{4} = 30\%$$

The minimum average return during the year for Venkat would be 30%.

Hence, the correct answer is option A.

[VIEW SOLUTION](#)

10. B

For the overall average to be 35%, there are two possible cases as shown below in the table,

Case 1: A gets double the expected results, and D gets one and a half times the expected results.

Case 2: C gets double the expected results, and A gets one and a half times the expected results.

		A	B	C	D	Average
Before Results		20	10	30	40	25
After Results	Case-1	40	10	30	60	35
	Case-2	30	10	60	40	35

In both cases, we can see that A must announce extraordinary results, and the results of B are as expected. But we can see that D has expected results in one case and had unexpected results in the other. Hence, statements 1 and 4 need not be necessarily true, and 2 and 3 must be true.

Hence, option B is the correct answer.

[VIEW SOLUTION](#)

100+ CAT Quant Questions



11. **B**

Cream would be undoubtedly the most profitable as maximum amount of white paint is used in it and white is the cheapest out of all other paints. Hence option B.

[VIDEO SOLUTION](#)

12. **B**

AVOCADO paint can be manufactured by adding orange and pink in equal quantity. 0.5 ltr of orange would cost 11 and the cheapest way to make pink would be by mixing white and red, so the cost for 0.5 ltr of pink comes out to be 8.75. So total cost becomes $11 + 8.75 = 19.75$ Rs. which is the cheapest.

[VIDEO SOLUTION](#)

13. **D**

WASHEDORANGE can be made by mixing orange and white, orange can be made by mixing equal quantities red and yellow which would be $\frac{1}{2}$ of white quantity. Thus RED, YELLOW, and WHITE in the ratio 1:1:2 are needed.

[VIDEO SOLUTION](#)

14. **B**

According to given conditions Bashir started with 500Rs Chirag should have $700 + \text{something}$, which when subtracted from 500 get money which Deepak has. Also Ashok should have < 1500 and his money should be 3 times that of deepak. Thus there's only 1 possibility that Ashok has 1200rs and deepak has 400rs. So the costliest item that deepak can buy is shawl worth 400 rs.

[VIEW SOLUTION](#)

15. **B**

According to given conditions, for 10 kg it will take 2 hrs/10kg and total cost for 80 kg to be 160rs. for 20 kg it will take 5 hrs/20kg and total cost for 80 kg to be 200rs, for 40 kg it will take 10 hrs/40kg and total cost for 80 kg to be 200rs. So the cheapest is 160 rs.

[VIEW SOLUTION](#)

100+ CAT DILR Questions



16. **C**

As it is mentioned that difference in number of books is more than one

Hence option B and D are cancelled.

Now total amount is = $620 = 17M + 19P + 23C$

we can check by putting values from option A and C.

Option C satisfies the above eq., Hence our answer will be C

[VIEW SOLUTION](#)

17. D

	M1	M2	M3	M4	M5	M6	M7	M8
LIKE	F	S	G	M4	D	F	G	S
	S	D					M	G
DISLIKE	X G	X F	X S	X D	X S	XS	XF	XM
					X M	XM		

Looking at each option and the table. Only Option D satisfies the given condition that each selected person shares a liking with at least one of the other three selected. The selected persons must also hate at least one of the likings of any of the other three persons selected.

[VIEW SOLUTION](#)

18. C

The points would be maximum when the person carries 1 management, 2 fiction, 2 maths and 5 physics book.

Hence total points $4+2+6+10 = 22$.

Hence option c.

[VIEW SOLUTION](#)

19. C

Correct arrangement would be

NAME	JACK	Mohan	Shalini	Suman	Rita
ENCLOSURES	Y	X	Q	P	Z
ANIMALS	BEAR	LION	LION	BISON	BISON
	PANTHER	PANTHER	BEAR	DEER	DEER

Hence option C.

[VIEW SOLUTION](#)

20. B

According to given condition we find average costs for products bought.

If D is bought then - $D + 2*B$. So we spend $40+180 = 220$ Rs for 3000 points. Hence, cost per 1000 points is 73.33

If A is bought then - $A+C$. So we spend $110 + 70 = 180$ for 2000 points. Hence, cost per 1000 points is 90

If E is bought then - $E+2*D+4*B+B$. So we spend $45 + 80 + 360 + 90 = 575$ for 8000 points. Hence, cost per 1000 points is 71.875.

If B is bought then I spend Rs 90 per 1000 points.

If C is bought then I spend Rs 70 per 1000 points.

To maximise points we need to select the item that costs the least per 1000 points. Hence, C costs the least per 1000 points. Hence, we should buy as many C items as possible.

Maximum C that can be bought is $[1000/70] = 14$ items and Rs 20 would be left over. However, the unspent

money would attract a penalty of 30000 points. Hence, instead of buying 14 C items, we should buy 13 C items and 1 B item so that total money spent is 1000.

[VIEW SOLUTION](#)

100+ CAT VARC Questions



21. **E**

A team which has K should have L also.

Since L is there in the team, N and U should not be there in the team. Since U is not there in the team, S and W should not be there in the team.

So, the team will have K, L, one of P and R and one of M or Q.

So, the team size will be 4.

[VIDEO SOLUTION](#)

22. **A**

If L is in the team, the team should include K also. The team should have one among P, R and S and one among M and Q.

So, the team size cannot be 3 if L is in the team.

[VIDEO SOLUTION](#)

23. **C**

Out of P, R and S only 1 can be in the team. If S is there, U and W will also be there. So, P and R should not be in the team for its size to be maximum.

Out of M and Q, only 1 can be there.

If L is there in the team, N and U cannot be in the team.

If L is not there in the team, then K is also not there in the team but N and U can be in the team.

So, the maximum team size is 5 consisting of S, U, W, (M or Q), N.

So, M can be a member of team size 5.

[VIDEO SOLUTION](#)

24. **E**

Since N is in the team, L and K cannot be in the team.

The team can have one of M and Q. So, 2 ways of selection.

If the team has S, then it should have U and W as well.

If the team has no S, then it should have one of P or R.

So, the number of ways of forming the team is $2 \times (1+2) = 6$ ways

[VIDEO SOLUTION](#)

25. **D**

Out of P, R and S only 1 can be in the team. If S is there, U and W will also be there. So, P and R should not be in the team for its size to be maximum.

Out of M and Q, only 1 can be there.

If L is there in the team, N and U cannot be in the team.

If L is not there in the team, then K is also not there in the team but N and U can be in the team.

So, the maximum team size is 5 consisting of S, U, W, (M or Q), N.

 VIDEO SOLUTION

CAT Expected Questions PDF

cracku



26. C

David and Peter cannot be selected together. So, option d) is not possible.

David and Fiza have to be selected together. So, option b) is ruled out.

Shyam and Rahim have to be selected together. So, option a) is also ruled out.

Option c) is the correct answer

 VIDEO SOLUTION

27. B

Shyam and Rahim have to be selected together. This rules out options a) and c). Ram will be in the group only if Peter is also there. This rules out option d). Option b) is a feasible group

 VIDEO SOLUTION

28. D

If Ram is selected, Peter also has to be selected. If Kavita is selected, David also has to be selected. But, Peter and David cannot be selected together. So, option a) is false.

If both the women are selected, David has to be selected. If David is selected, Peter cannot be selected. So, Ram also cannot be selected. Shyam and Rahim have to be selected together but there is room for only one more person. So, option b) cannot be true.

If David is selected in the group, Fiza has to be there. So, David cannot be in a group of all men. But, Ram and Shyam cannot be selected together. So, a group of all men is not possible. So option c is false.

 VIDEO SOLUTION

29. B

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

This is because Chetan buys for every fall in price and sells for every rise in price. But the fluctuation in share price is constant as it starts from 100 and closes at 110 on day 5. So, in total Chetan will always sell 10 shares in all 5 days combined.

As Chetan has sold 10 shares, he'll get $110 \times 10 = 1100$ cash because of it in every case. Also, Chetan earns Rs. 200 in every case because of buying at low and selling at high. So total cash Chetan will always have after 5 days = $1100 + 200 = 1300$.

The question asks about the case where Michael ended up with Rs 1300 cash less than Chetan.

As Chetan will have exactly 1300 cash in every case; we have to look for the cases where Michael does not make any profit and also does not have any sold shares at the end of 5 days.

This is possible in 4 cases ie. 3,4,5 and 9.

In each of these cases, the price fluctuates in the range 90-110 which does not allow Michael to buy or sell any shares.

Also, in all these cases the price of MCS is 100 at the end of day 4.

So the answer is 100.

 VIDEO SOLUTION

30. D

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

This is because Chetan buys for every fall in price and sells for every rise in price. But the fluctuation in share price is constant as it starts from 100 and closes at 110 on day 5. So, in total Chetan will always sell 10 shares in all 5 days combined.

As Chetan has sold 10 shares, he'll get $110 \times 10 = 1100$ cash because of it in every case. Also, Chetan earns Rs. 200 in every case because of buying at low and selling at high. So total cash Chetan will always have after 5 days = $1100 + 200 = 1300$.

The question asks about the case where there has been the maximum possible increase in the combined cash balance of Chetan and Michael at the end of the fifth day.

As we know chetan has 1300 cash in all the cases , so we have to maximize the case where Michael has the most cash.

Also, if we see clearly, the profit made by selling and buying is in hundreds while the cash received by selling the shares is much far in terms of cash.

As 10 shares sold give = $10 \times 110 = 1100$ cash. So we have to look at the case where Michael has sold most shares which is case 8.

In case 8 the price rises from 100 to 110 $\rightarrow$ 120 $\rightarrow$ 130 $\rightarrow$ 120 $\rightarrow$ 110

in this case 110(0 new shares) $\rightarrow$ 120(Michael sold 10 shares) $\rightarrow$ 130(Michael sold 10 shares) 120(Michael sold 10 shares) $\rightarrow$ 110(Michael does nothing) = 30 shares sold in total

Cash = 20 shares at 120 and 10 shares at 130 = $120 \times 20 + 130 \times 10 = 2400 + 1300 = 3700$

As Chetan has 1300 cash and Michael has 3700 cash;
total cash they have is : $3700 + 1300 = 5000$

 VIDEO SOLUTION



31.C

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

The question asks about the case where Chetan has sold 3 times and Michael sells only once.

Starting with Michael, for exactly one sell, the price should touch 120 only once as Michael sells the share only at price greater than 110.

if the price touches 120 twice or more, Michael will sell the share more than once which is not a desirable case.

Also, chetan has to sell thrice consecutively which is only possible if the share price is 90 at one instance and rises to 120 in straight 3 days.

This is only possible in case 2. Hence the price on 3rd day's end in case 2 is 110.

 VIDEO SOLUTION

32.E

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

This is because Chetan buys for every fall in price and sells for every rise in price. But the fluctuation in share price is constant as it starts from 100 and closes at 110 on day 5. So, in total chetan will always sell 10 shares in all 5 days combined.

As chetan has sold 10 shares , he'll get $110 \times 10 = 1100$ cash because of it in every case. Also chetan earns Rs. 200 in every case because of buying at low and selling at high. So total cash chetan will always have after 5 days = $1100 + 200 = 1300$.

The question asks about the case where Michael ended up with Rs 100 less cash than Chetan at the end of day 5.

So we have to look for the case where Michael has 1200 cash which is only possible when Michael has -10 number of shares in the end of day 5 and also has made profit of Rs. 100.

This is possible in case 7 as Michael has sold the shares at rs 120 but did not sell those shares as price never went below 90.

In the end Michael will have -10 shares at a price 110 which is Rs. 10 less than the price he sold at.

So he makes profit of 100 rs through it and will have the cash of 1100 through the sold shares.

Their difference in 7th case is $1300 - 1200 = 100$

Also, in this case they have same number of shares sold at the end of the 5th day. So E is the answer.

 VIDEO SOLUTION

33. A

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

To have 20 more shares than Chetan, Michael has to buy 10 shares which is case 1.

In case 1, the share price at the end of day 3 is 90.

 VIDEO SOLUTION

34. C

From the first condition a, assume the number of experts in labour = x, then the number of experts in each of the other three categories = 2x

Now, $x + 2x + 2x + 2x + 2x = 21 \Rightarrow x = 3$ and $2x = 6$

So the number of experts in labour = 3 and the number of experts in each of the three categories = 6

From condition d, if the number of experts of Astralaise = y, then the number of experts of Americas = $2(y-1)$,

the number of experts of Europe = $y-1$, the number of experts of Africa = $y-1$

Now, $y+y-1+y-1+2(y-1)=21 \Rightarrow 5y-4=21 \Rightarrow y=5$

Hence the table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour					3
Health					6
Population Studies					6
Refugee Location					6
Total	4	8	5	4	21

Using condition b, the number of labour experts of Africa = 0 and the number of labour experts of the rest of the continents will be 1 each.

Consider the column of Europe, since Europe sent at least 1 in each category, each category will have 1 expert. (Since the total sum is 4)

From the condition e, the population studies of Australasia will contain 2 experts. So all the other categories will have to take at 1 expert because the total sum is 5 and no value can be 0. The table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour	0	1	1	1	3
Health			1	1	6
Population Studies			2	1	6
Refugee Location			1	1	6
Total	4	8	5	4	21

Now, according to the condition c, the possible solutions are as given in the tables below:

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	2	2	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	1	3	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	2	2	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
Health	1	3	1	1	6
Population Studies	2	1	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

From the tables we can see that when there is only 1 population expert from America, there are 3 American Health Experts. Hence option C is not true.

[VIDEO SOLUTION](#)

35. D

From the first condition a, assume the number of experts in labour = x , then the number of experts in each of the other three categories = $2x$

Now, $x+2x+2x+2x+2x = 21 \Rightarrow x = 3$ and $2x = 6$

So the number of experts in labour = 3 and the number of experts in each of the three categories = 6

From condition d, if the number of experts of Astralasia = y , then the number of experts of Americas = $2(y-1)$, the number of experts of Europe = $y-1$, the number of experts of Africa = $y-1$

Now, $y+y-1+y-1+2(y-1)=21 \Rightarrow 5y-4=21 \Rightarrow y=5$

Hence the table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour					3
Health					6
Population Studies					6
Refugee Location					6
Total	4	8	5	4	21

Using condition b, the number of labour experts of Africa = 0 and the number of labour experts of the rest of the continents will be 1 each.

Consider the column of Europe, since Europe sent at least 1 in each category, each category will have 1 expert. (Since the total sum is 4)

From the condition e, the population studies of Australasia will contain 2 experts. So all the other categories will have to take at 1 expert because the total sum is 5 and no value can be 0. The table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour	0	1	1	1	3
Health			1	1	6
Population Studies			2	1	6
Refugee Location			1	1	6
Total	4	8	5	4	21

Now, according to the condition c, the possible solutions are as given in the tables below:

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	2	2	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	1	3	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	2	2	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
Health	1	3	1	1	6
Population Studies	2	1	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

From the above tables we can see that there was no possible case in which Africa and America had 1 expert each in population studies. Thus, statement D is false.

[VIDEO SOLUTION](#)



36. C

From the first condition a, assume the number of experts in labour = x , then the number of experts in each of the other three categories = $2x$

Now, $x+2x+2x+2x+2x = 21 \Rightarrow x = 3$ and $2x = 6$

So the number of experts in labour = 3 and the number of experts in each of the three categories = 6

From condition d, if the number of experts of Australasia = y , then the number of experts of Americas = $2(y-1)$, the number of experts of Europe = $y-1$, the number of experts of Africa = $y-1$

Now, $y+y-1+y-1+2(y-1)=21 \Rightarrow 5y-4=21 \Rightarrow y=5$

Hence the table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour					3
Health					6
Population Studies					6
Refugee Location					6
Total	4	8	5	4	21

Using condition b, the number of labour experts of Africa = 0 and the number of labour experts of the rest of the continents will be 1 each.

Consider the column of Europe, since Europe sent at least 1 in each category, each category will have 1 expert. (Since the total sum is 4)

From the condition e, the population studies of Australasia will contain 2 experts. So all the other categories will have to take at 1 expert because the total sum is 5 and no value can be 0. The table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour	0	1	1	1	3
Health			1	1	6
Population Studies			2	1	6
Refugee Location			1	1	6
Total	4	8	5	4	21

Now, according to the condition c, the possible solutions are as given in the tables below:

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	2	2	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	1	3	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	2	2	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
Health	1	3	1	1	6
Population Studies	2	1	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

If an American expert in refugee relocation, was the first keynote speaker in the conference. Then apart from him there is atleast 1 and atmost 2 american expert in refugee relocation. Hence option C.

 VIDEO SOLUTION

37. D

From the first condition a, assume the number of experts in labour = x, then the number of experts in each of the other three categories = 2x

Now, $x + 2x + 2x + 2x + 2x = 21 \Rightarrow x = 3$ and $2x = 6$

So the number of experts in labour = 3 and the number of experts in each of the three categories = 6

From condition d, if the number of experts of Astralasia = y, then the number of experts of Americas = $2(y-1)$, the number of experts of Europe = $y-1$, the number of experts of Africa = $y-1$

Now, $y + y-1 + y-1 + 2(y-1) = 21 \Rightarrow 5y-4=21 \Rightarrow y=5$

Hence the table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour					3
Health					6
Population Studies					6
Refugee Location					6
Total	4	8	5	4	21

Using condition b, the number of labour experts of Africa = 0 and the number of labour experts of the rest of the continents will be 1 each.

Consider the column of Europe, since Europe sent at least 1 in each category, each category will have 1 expert. (Since the total sum is 4)

From the condition e, the population studies of Australasia will contain 2 experts. So all the other categories will have to take at 1 expert because the total sum is 5 and no value can be 0. The table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour	0	1	1	1	3
Health			1	1	6
Population Studies			2	1	6
Refugee Location			1	1	6
Total	4	8	5	4	21

Now, according to the condition c, the possible solutions are as given in the tables below:

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	2	2	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	1	3	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	2	2	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
Health	1	3	1	1	6
Population Studies	2	1	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

We can see that there are 2 possibilities for number of experts in refugee relocation from Africa. Hence it cannot be determined.

 VIDEO SOLUTION

38. C

From statement 1 and 2, Each team speaks exactly four languages. English and Chinese are spoken by all three teams, Basque and French by exactly two teams and the other languages by exactly one team, multiple options are possible.

In the following tables: A, B, C can be any team among Team 1, Team 2, Team 3.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
French	French	Dutch

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Dutch
Arabic	French	French

From the data given in the question, the person who speaks Arabic also speaks French. Thus the only option possible is 'Table 2'.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

According to statement 4, "Paula and Sally are together in exactly two teams."

Sally knows Basque, thus, she will be in group A and B, with Paula.

According to statement 5, Robert(Arabic) is in Team 1 and Quentin(Dutch) is in Team 3.

Thus, Group C is Team 1 and Group A is Team 3.

Team 3		Team 2		Team 1	
English	Paula, Sally, Quentin	English	Paula, Sally	English	Paula
Chinese	Paula	Chinese	Paula, Terence	Chinese	Paula, Terence
Basque	Sally	Basque	Sally	Arabic	Terence, Robert
Dutch	Quentin	French	Terence	French	Robert

From the table, the correct option is C.

 VIDEO SOLUTION

39. C

From statement 1 and 2, Each team speaks exactly four languages. English and Chinese are spoken by all three teams, Basque and French by exactly two teams and the other languages by exactly one team, multiple options are possible.

In the following tables: A, B, C can be any team among Team 1, Team 2, Team 3.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
French	French	Dutch

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Dutch
Arabic	French	French

From the data given in the question, the person who speaks Arabic also speaks French. Thus the only option possible is 'Table 2'.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

According to statement 4, "Paula and Sally are together in exactly two teams."

Sally knows Basque, thus, she will be in group A and B, with Paula.

According to statement 5, Robert(Arabic) is in Team 1 and Quentin(Dutch) is in Team 3.

Thus, Group C is Team 1 and Group A is Team 3.

Team 3		Team 2		Team 1	
English	Paula, Sally, Quentin	English	Paula, Sally	English	Paula
Chinese	Paula	Chinese	Paula, Terence	Chinese	Paula, Terence
Basque	Sally	Basque	Sally	Arabic	Terence, Robert
Dutch	Quentin	French	Terence	French	Robert

From the table, the correct option is C.

 VIDEO SOLUTION

40. C

From statement 1 and 2, Each team speaks exactly four languages. English and Chinese are spoken by all three teams, Basque and French by exactly two teams and the other languages by exactly one team, multiple options are possible.

In the following tables: A, B, C can be any team among Team 1, Team 2, Team 3.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
French	French	Dutch

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Dutch
Arabic	French	French

From the data given in the question, the person who speaks Arabic also speaks French. Thus the only option possible is 'Table 2'.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

According to statement 4, "Paula and Sally are together in exactly two teams."

Sally knows Basque, thus, she will be in group A and B, with Paula.

According to statement 5, Robert(Arabic) is in Team 1 and Quentin(Dutch) is in Team 3.

Thus, Group C is Team 1 and Group A is Team 3.

Team 3		Team 2		Team 1	
English	Paula, Sally, Quentin	English	Paula, Sally	English	Paula
Chinese	Paula	Chinese	Paula, Terence	Chinese	Paula, Terence
Basque	Sally	Basque	Sally	Arabic	Terence, Robert
Dutch	Quentin	French	Terence	French	Robert

From the table, the correct option is C.

 VIDEO SOLUTION



CAT Mock Test

Mocks designed exactly like the actual CAT




41. A

From statement 1 and 2, Each team speaks exactly four languages. English and Chinese are spoken by all three teams, Basque and French by exactly two teams and the other languages by exactly one team, multiple options are possible.

In the following tables: A, B, C can be any team among Team 1, Team 2, Team 3.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
French	French	Dutch

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Dutch
Arabic	French	French

From the data given in the question, the person who speaks Arabic also speaks French. Thus the only option possible is 'Table 2'.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

According to statement 4, "Paula and Sally are together in exactly two teams."

Sally knows Basque, thus, she will be in group A and B, with Paula.

According to statement 5, Robert(Arabic) is in Team 1 and Quentin(Dutch) is in Team 3.

Thus, Group C is Team 1 and Group A is Team 3.

Team 3		Team 2		Team 1	
English	Paula, Sally, Quentin	English	Paula, Sally	English	Paula
Chinese	Paula	Chinese	Paula, Terence	Chinese	Paula, Terence
Basque	Sally	Basque	Sally	Arabic	Terence, Robert
Dutch	Quentin	French	Terence	French	Robert

From the table, the correct option is A.

 VIDEO SOLUTION